4/3/26 | Jack Hasker, Emmett Souder, Eddie Kong | Networks Project Progress Report

Problem Statement: Resilience and Critical Nodes in the ERCOT Grid

Winter Storm Uri in 2021 exposed catastrophic vulnerabilities in the Electric Reliability Council of Texas (ERCOT) grid's structure. For this project, we analyze the ERCOT transmission network to identify its most critical infrastructure, which we define as nodes whose failure would disconnect the largest, most population-dense areas. We plan on using advanced network analysis to move beyond simple connectivity to derive a data-backed ranking of critical infrastructure that can be used to prioritize the most beneficial cost efficient upgrades for the ERCOT grid. Our analysis culminates into a hypothetical construction of an optimal, resilient network designed from the ground up to maximize robustness and reliability against hypothetical extreme weather events.

Revised Plan:

After some initial research, we realized that using our model to predict energy prices is something that is way out of the scope. It would be a project on its own. Instead we have decided to discuss the creation of an optimal network that we build from the ground up based on various constraints. Some ideas are cost minimization, resilience maximizing, highest growth potential, or a combination of some. We plan on continuing the same path for the construction and analysis of the network.

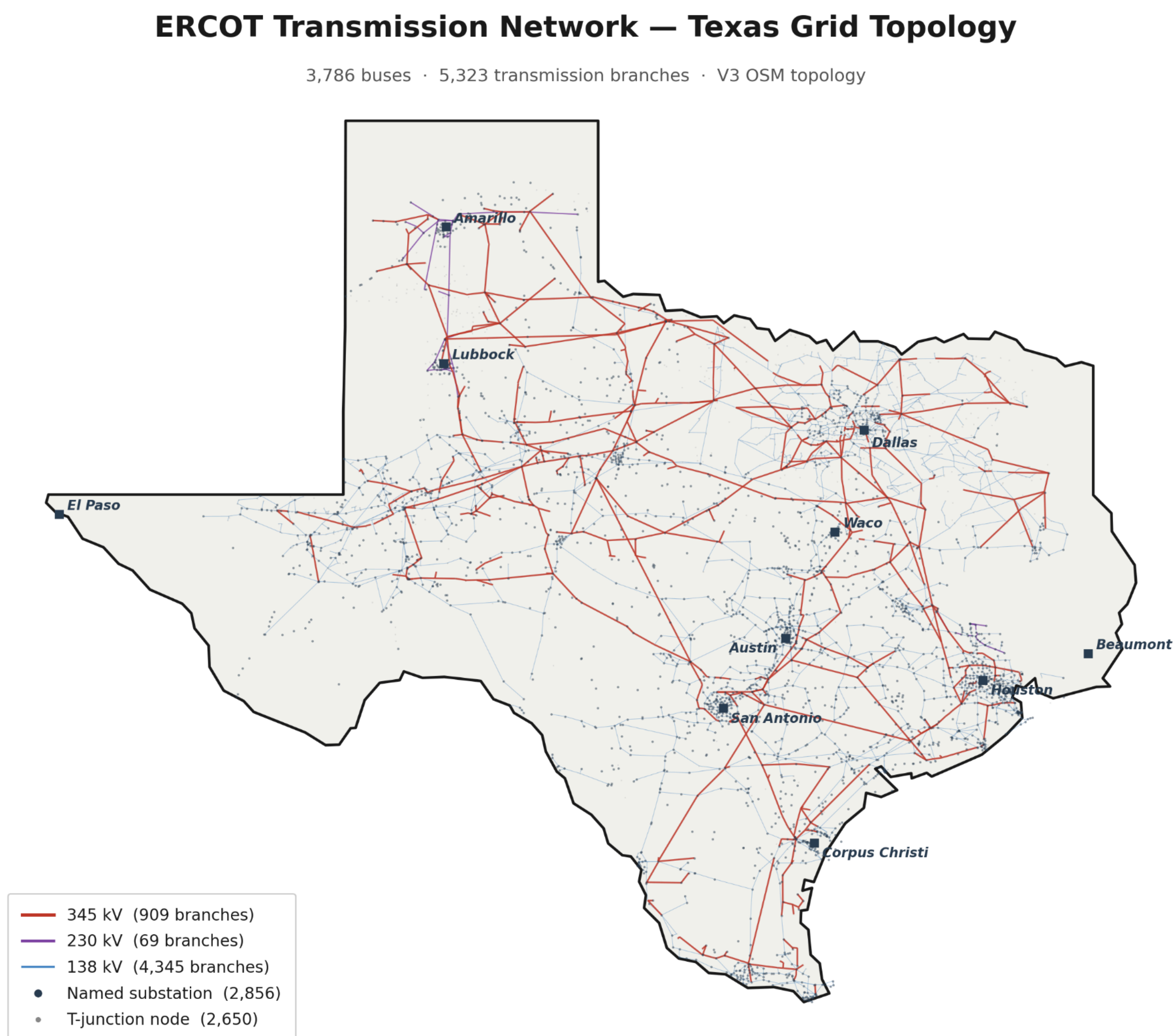
Progress Thus Far:

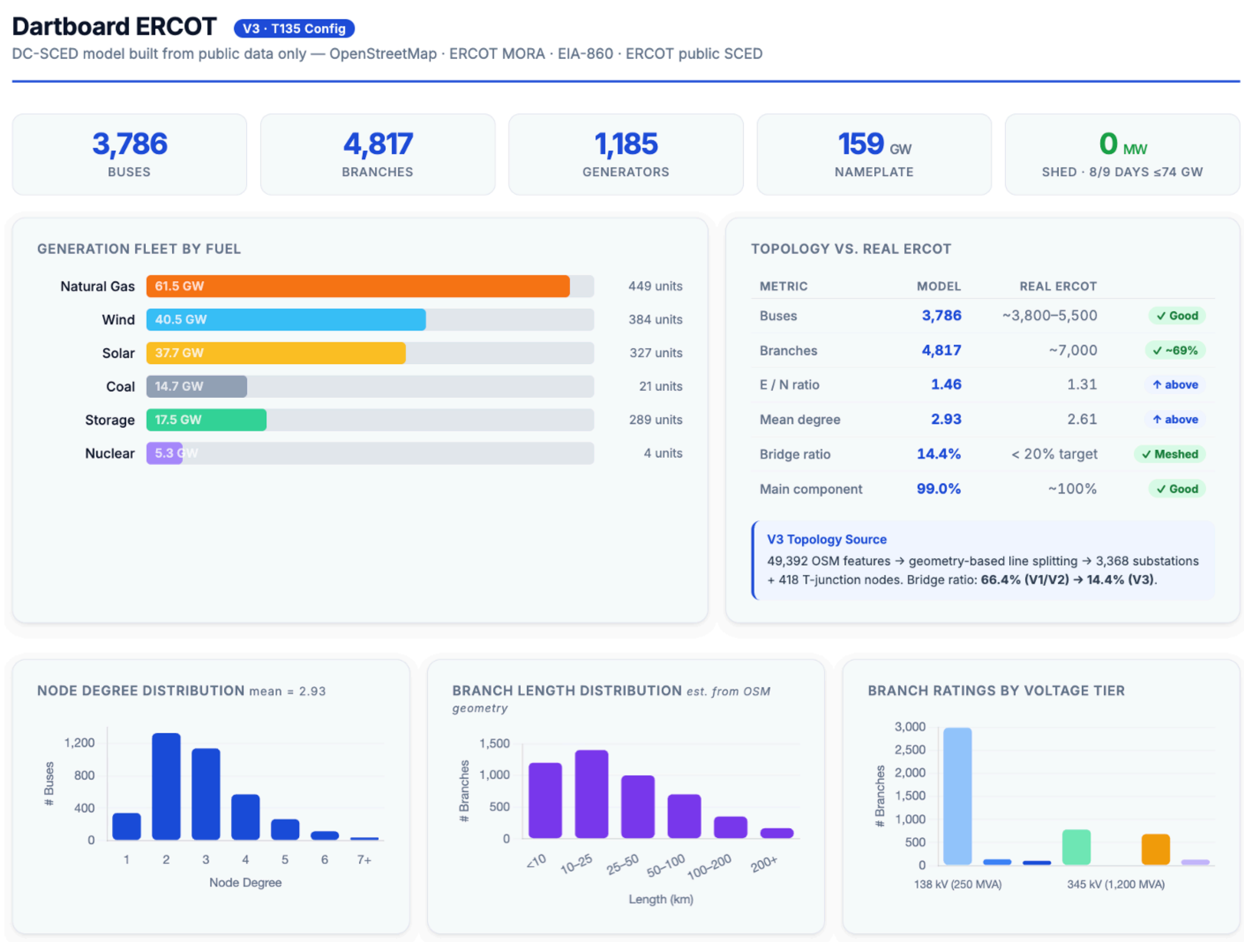
Below are updates based on the sections of what we have accomplished so far. A majority of the updates are in the network construction because the data was extremely messy, and we have to make a network before we make any analysis.

Constructing the ERCOT Grid:

The network model was constructed by merging multiple public sources of infrastructure data (data) and running a security constrained economic dispatch (SCED) algorithm to tune line rating to produce reasonable results. The basic underlying principles of the network are geospatial mapping, infrastructure data, and population data. To start, the network is constructed to match the geographic location of the substations and generation plants. We used a Leaflet.js mapping library to plot precise latitude and longitude coordinates for each component. Once we had the locations, we had to assign the capacities of the nodes. This is where the infrastructure type was implemented. We cross reference the type of node, generation vs. substation, and its location. Each generation plant then has a production capacity and we also added the type of plant that they were (solar, coal, etc.). Once the generators were classified, the next challenge was implementing a reasonable load (demand) distribution. We decided to use county level census data, as higher granularity would lead to inaccurate results: since we already simplified some aspects of the model (e.g. ignoring distribution level lines at 69kv and lower), we adjusted the resolution of other parameters correspondingly. Specifically, fine concentrations of a population's power demand can't be properly met when we do not represent the lower level distribution network, so it's better to use a coarser, averaged-out demand as the higher level network. The last hurdle was translating the messy Open Street Map geographic line data into edges and nodes (or busses and branches, in power systems parlance). We first had to extract it from the open source file, and then clean up all of the errors like incomplete line, dead ends, disconnected points. A lot of this required visual correction, using Claude Code to implement traversal algorithms to connect lines and edges and then debugging visibly wrong aspects. See appendix for details. Open Street Map data has voltage tags, but we had to estimate the capacity of the lines. As mentioned earlier, this was done using a SCED model and a series of upgrades until the topology produced reasonable results. As disclosed at office hours, this work overlaps with Emmett's independent research project, but his IW didn't delve into any networks methods or analysis and our project is based on changing that markets model (the SCED model) to a more useful model for our project (which we've done), and running analysis on this new model. This class project involves more robust network construction, whole graph analysis, and our own new methodologies and experiments.

After constructing the network we synthesized some basic characteristics listed below:





Analysis on Existing Grid:

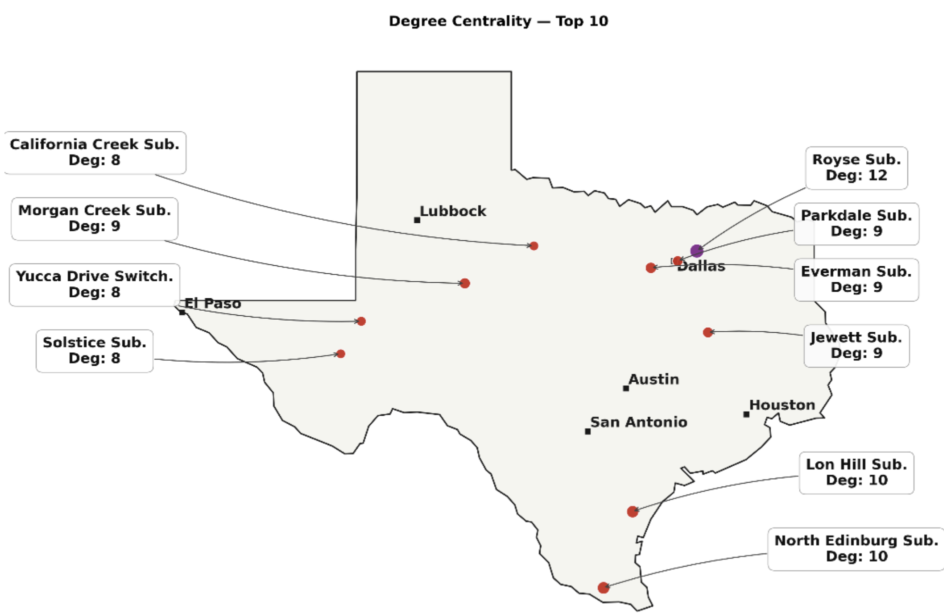
Once we created the graph, albeit a little more simplified, we can run analysis on what makes the Texas grid unique. Unlike other major North American interconnects, ERCOT operates as a largely isolated electrical island, meaning it does not have extensive direct-current (DC) connections to the Eastern and Western grids. This means the grid has to be self-sustaining, so it more or less uses all the electricity it produces, as it is harder to export and import electricity from other interconnects.

To address the need for a truly resilient network, the goal of this analysis is to identify critical nodes, those whose failure results in the largest, population-weighted disconnections. We started with an initial analysis of using degree centrality. We calculated the degree centrality for all nodes by tallying the total number of transmission line edges connected to each bus. Since the network edges are described in the code with their endpoints, this was a straightforward tally.

1. Royse Substation: Degree = 12 (Node ID: 2489)

2. North Edinburg Substation: Degree = 10 (Node ID: 2384)
3. Lon Hill Substation: Degree = 10 (Node ID: 2398)
4. Everman Substation: Degree = 9 (Node ID: 2497)
5. Morgan Creek Substation: Degree = 9 (Node ID: 2504)
6. Jewett Substation: Degree = 9 (Node ID: 2719)
7. Parkdale Substation: Degree = 9 (Node ID: 2798)
8. Unnamed 345 kV junction near Lubbock: Degree = 9 (Node ID: 3521)
9. Yucca Drive Switching Station: Degree = 8 (Node ID: 78)
10. California Creek Substation: Degree = 8 (Node ID: 386)

After we found the top nodes, we noticed a pattern. All of these nodes are substations, which means they are connected to low voltage lines. In a literal sense these nodes are highly connected, but their reach is only local. These nodes are only important to their local surroundings rather than the grid as a whole. We wanted to find larger hubs and therefore needed to change our search mechanism. Below is a picture of all of the nodes that have the highest degree. There is a slight concentration around Dallas, but overall there are no geographically significant patterns. These nodes are important but did not meet our goal of finding critical infrastructure.



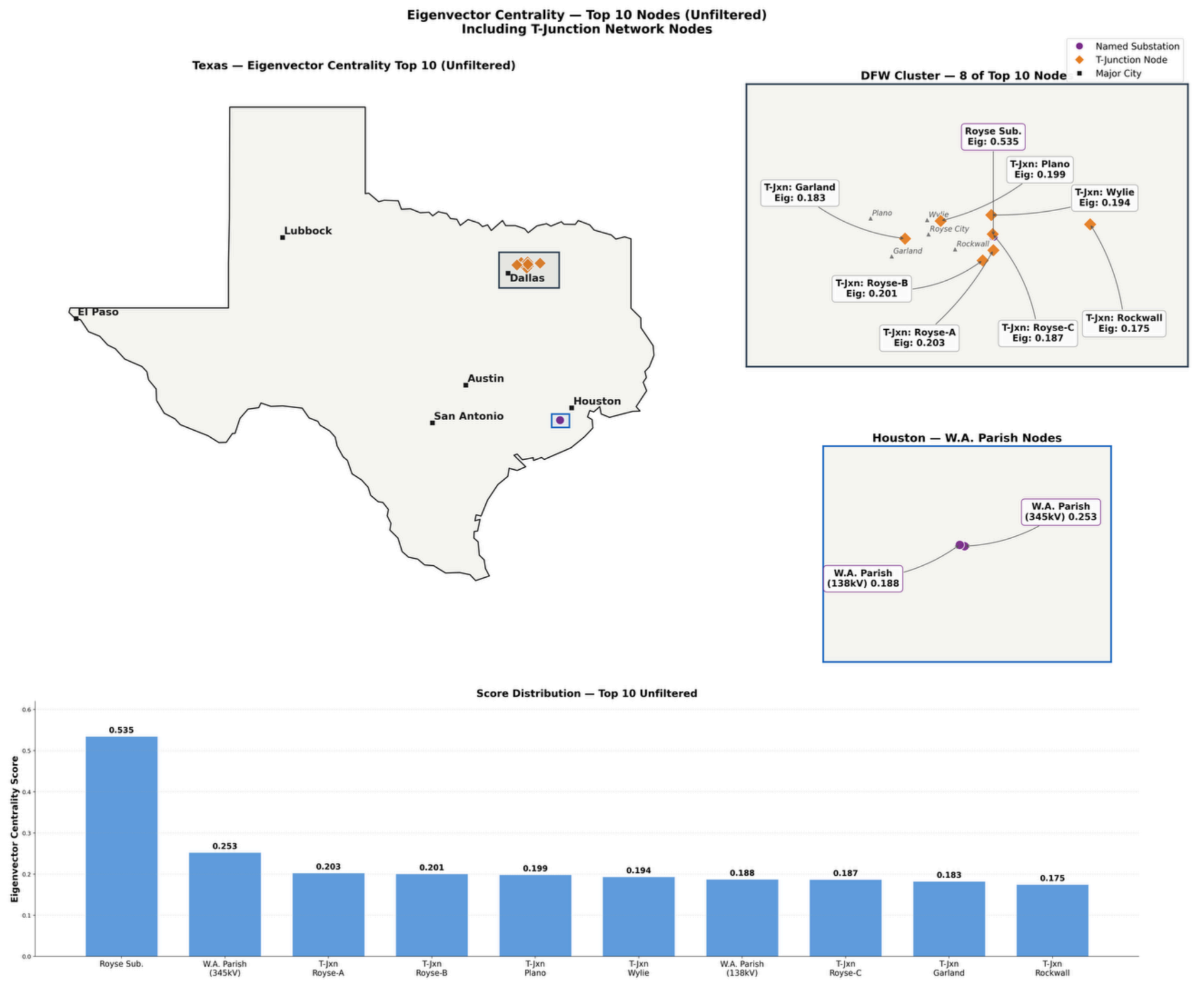
Our next analysis involved using Eigenvector Centrality. This centrality measure takes into account how important each node is to other important nodes. It assigns higher mathematical scores to substations that are topologically adjacent to the structural core of the network. In order to calculate this, we need to derive the network's adjacency matrix. In this matrix, a value of 1 in element *ij* represents the existence of a transmission line between node *i* and node *j*. The eigenvector centrality of a node is defined as being proportional to the sum of the centralities of its direct neighbors.

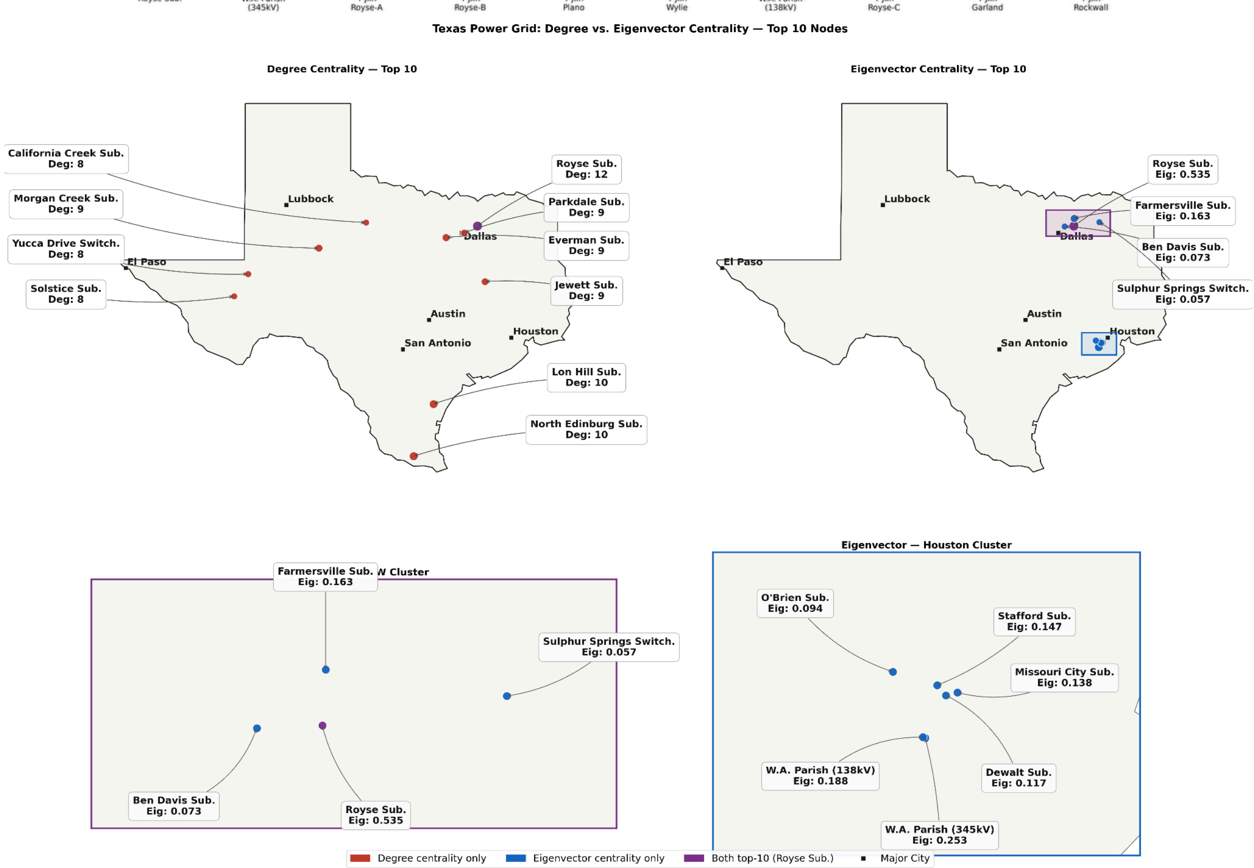
$$x_i = \frac{1}{\lambda} \sum_{j=1}^{N} A_{i,j} x_j$$

In matrix notation, this forms the classic eigenvalue equation:

$$A\mathbf{x} = \lambda \mathbf{x}$$

We decided to compute the eigenvector centrality from scratch. To do this, we used the power iteration algorithm with a convergence tolerance of 10^{-8}. As done in class, we started from a uniform vector and repeatedly updated each node's score as the sum of its neighbors' scores, then normalized by the L2 norm. The graph has three edge arrays, EDGES_HIGH (345 kV), EDGES_MID_HI (230 kV), and EDGES_MID_LO (138 kV), totalling 4,268 nodes with at least one connection and 5,323 transmission line edges. The corresponding adjacency matrix was very sparse with 99.94% of the elements being zero. In the final eigenvector given by the algorithm, a large majority of the top 10 nodes were unnamed T-junctions. These nodes, which we call T-junctions, are synthetic nodes critical for accurately representing the grid's topology. In the real world, a transmission line often branches off another line without passing through a substation. For the graph to correctly model this branching behavior, which differs significantly from a direct node-to-node connection, we created T-junction nodes to be placed at these branching points to ensure the topology was correctly represented. A detailed description of the implementation of the T-junction modeling is available in the Appendix. In total, we created 418 synthetic T-junction nodes. Below are the eigenvalue centrality scores.


Eigenvector Centrality — Top 10 Nodes (Unfiltered)
Including T-Junction Network Nodes
Texas — Eigenvector Centrality Top 10 (Unfiltered)
Lubbock
El Paso
Dallas
Austin
San Antonio
Houston
DFW Cluster — 8 of Top 10 Node
Named Substation
T-Junction Node
Major City
Royse Sub.
Eig: 0.535
T-Jxn: Plano
Eig: 0.199
T-Jxn: Garland
Eig: 0.183
T-Jxn: Wylie
Eig: 0.194
Plano
Wylie
Royse City
Garland
Rockwall
T-Jxn: Royse-B
Eig: 0.201
T-Jxn: Royse-A
Eig: 0.203
T-Jxn: Royse-C
Eig: 0.187
T-Jxn: Rockwall
Eig: 0.175
Houston — W.A. Parish Nodes
W.A. Parish
(345kV) 0.253
W.A. Parish
(138kV) 0.188
Score Distribution — Top 10 Unfiltered
Eigenvector Centrality Score
0.535
0.253
0.203
0.201
0.199
0.194
0.188
0.187
0.183
0.175
Royse Sub.
W.A. Parish (345kV)
T-Jxn Royse-A
T-Jxn Royse-B
T-Jxn Plano
T-Jxn Wylie
W.A. Parish (138kV)
T-Jxn Royse-C
T-Jxn Garland
T-Jxn Rockwall


Texas Power Grid: Degree vs. Eigenvector Centrality — Top 10 Nodes
Degree Centrality — Top 10
California Creek Sub.
Deg: 8
Morgan Creek Sub.
Deg: 9
Yucca Drive Switch.
Deg: 8
Solstice Sub.
Deg: 8
Royse Sub.
Deg: 12
Parkdale Sub.
Deg: 9
Everman Sub.
Deg: 9
Jewett Sub.
Deg: 9
Lon Hill Sub.
Deg: 10
North Edinburg Sub.
Deg: 10
Lubbock
El Paso
Dallas
Austin
San Antonio
Houston
Eigenvector Centrality — Top 10
Royse Sub.
Eig: 0.535
Farmersville Sub.
Eig: 0.163
Ben Davis Sub.
Eig: 0.073
Sulphur Springs Switch.
Eig: 0.057
W Cluster
Farmersville Sub.
Eig: 0.163
Sulphur Springs Switch.
Eig: 0.057
Ben Davis Sub.
Eig: 0.073
Royse Sub.
Eig: 0.535
Eigenvector — Houston Cluster
O'Brien Sub.
Eig: 0.094
Stafford Sub.
Eig: 0.147
Missouri City Sub.
Eig: 0.138
W.A. Parish (138kV)
Eig: 0.188
Dewalt Sub.
Eig: 0.117
W.A. Parish (345kV)
Eig: 0.253
Degree centrality only
Eigenvector centrality only
Both top-10 (Royse Sub.)
Major City

It is easy to see how these nodes are very geographically patterned. They match the most populated areas in Texas, which are the largest demand centers. Interestingly, the Royse substation is still the number one node in both of these instances, but none of the other 9 high degree nodes make it in the eigenvalue list. These nodes are located in dense parts of the grid, so they accumulate high eigenvector scores due to their connectivity to large populations. This is the likely explanation for the two high-scoring W.A. Parish 345 kV and 138 kV nodes. However, since these T-junction are artificially created, we also ran an eigenvalue search that ignored the T-junctions. Interestingly, these also remained around the large population centers but were slightly more spread out.

Great! We found the most important nodes in Texas's Grid. Well, only by one metric. These nodes are important because there is (likely, by our heuristic) the largest quantity of electricity flowing through them, but they are not *necessarily* critical. They may not be vital to supplying power in the area: due to their well-connectedness, there are redundancies in place to provide continuous power if these nodes were to fail. This is exactly what eigenvalue searches are meant to do. However, we are not looking for these "most important" nodes– we are looking for the most critical nodes, which in particular means that a large number of people rely on them *uniquely* for their source of power. In other words, if that particular node goes down, there would be a large number of people without connection to the grid. These would be prime locations to upgrade the robustness of the current network.

Progress Planned for the Final Report:

We still need to find nodes that are most critical to the Texas grid from a vulnerability standpoint. Our ideas are using some of the partitioning algorithms that we learned in class. For example, we found an algorithm called Articulation Point Analysis which would let us know which nodes would disconnect the most nodes. We plan to make the analysis deeper by weighing the scores by population and testing other heuristics. A node is more critical if it disconnects Houston than if it disconnects a rural west Texas town. This also made us curious about specific lines: would lines around these critical nodes also give high edge betweenness scores? We would evaluate edges similarly to our node centrality analyses. Other algorithms like Unsupervised Link Prediction could also point to ways to make the grid more robust by finding the strongest future connections that would serve as a proxy for strengthening weak points.

After these algorithms, we aim to have an extensive list of nodes and lines that are important *and* critical to the grid. We plan to cross reference our nodes with the nodes that have been critical or important in the past using publicly available outage and upgrade data like FERC OE-417 which reports utilities outages affecting ≥50,000 customers and/or lasting >1hr, the EIA Form 861, which publishes the annual reliability stats (SAIDI/SAIFI) by Texas utilities (Oncor, CenterPoint, AEP, etc.), and lastly, ERCOT post-event reports which detail the aftermaths from storms Uri (Feb 2021) and Elliott (Dec 2022). All of this information will be consolidated into a cost benefit analysis for the best lines that would have the most impact for upgrading the grid's reliability and robustness.

These studies will conclude the reality part of the project, which was part 2 of our outline. The final part of our project would be to construct the perfect Texas grid based solely on the load distribution that we created based on population data. We will use a variety of optimality measures like cost, robustness, connectivity, and growth potential. These are just preliminary ideas and will likely evolve in office hours meetings. We are open to any suggestions that you think would be fun or interesting to construct a network around.

Appendix:

Extracting OSM topology and processing it into a usable graph has been done before manually, but we automated the process with Claude Code. Code available on request. The full topology construction specs:

Topology Construction Method

The data ingestion pipeline builds a DC-SCED-ready network graph of ERCOT entirely from public data.It follows a PyPSA-Eur-inspired approach: geometry-based line splitting rather than endpoint-only matching. The key insight is that OSM coordinates are GPS-precise. The data is accurate, so the pipeline just needs to not be lossy.

Stage 1: Data Ingestion (generate_visualizer.py)

Inputs:

- texas_substations.geojson — 5,786 OSM substations (all voltages, not just HV)

- texas_hv_lines.geojson — 22,913 HV transmission line geometries from OSM

- texas_plants.geojson — OSM power plant locations

- ercot_zones.geojson — ERCOT load zone polygons (4 zones)

- texas_matched_substations_v6.csv — ERCOT-to-OSM substation matching

- EIA-860 generator data

Stage 2: Node Set Construction

1. Load all OSM substations inside ERCOT territory (point-in-polygon against zone boundaries). Includes all voltages (69 kV, 115 kV, untagged) — not just HV — because these serve as transformer endpoints.

2. Intra-complex dedup (200 m): Multiple OSM features at the same physical substation complex (same voltage tier) are collapsed into one node, keeping the named/highest-ID representative.

3. Cross-voltage merge (300 m): Substations at different voltage tiers within 300 m are merged into a single bus (keep the higher-kV node). Real transformers are local equipment at one physical location, not branches between distant nodes.

Stage 3: Geometry-Based Line Splitting (the core algorithm)

For each HV line feature:

1. Walk the full coordinate geometry segment by segment

2. For each segment, search nearby substations (spatial grid, 300 m buffer)

3. Compute perpendicular distance from each substation to the line segment

4. If distance < 300 m, record a "split hit" with position along the line (0.0 to 1.0)

5. Also snap line endpoints to nearest substation (dynamic radius: 500 m for ≥345 kV, 400 m for lower)

6. Sort all hits by position along the line

7. Create edges between consecutive substations along the route

This means a single 200 km OSM line that passes near 5 substations produces 4 edges — not just one endpoint-to-endpoint edge. This is what makes V3 meshed (14.4% bridge ratio vs 66.4% in V1/V2).

Stage 4: Junction Handling

Lines where one or both endpoints don't snap to any substation:

1. Cluster unsnapped endpoints within 150 m

2. Clusters near substations snap to them

3. Clusters with degree ≥ 3 become T-junction split nodes (synthetic nodes)

4. BFS contraction through degree-2 junction chains to connect stop nodes (substations + split points)

5. Partial connection resolution: 1-hit lines (one end snapped, one didn't) get a wider 1 km snap for the unsnapped end

Stage 5: Dead-End Pruning

Iteratively remove degree-1 nodes that aren't substations (dangling stubs that don't connect to anything useful).

Stage 6: Branch Table (build_osm_branch_table_v3.py)

Reads the NODES and EDGES from the visualizer HTML and builds branch.csv:

1. Impedance: $X_{pu} = x_{ohm\_per\_km} * route\_distance\_km / (kV^2 / 100)$. Route distance from actual OSM geometry, not straight-line haversine.

2. Ratings by voltage tier:

   - 138 kV: 250 MVA (single-circuit Drake ACSR — physically correct, no compensating error)

   - 230 kV: 400 MVA

   - 345 kV: 1,200 MVA base, upgraded to 2,400 MVA (double-circuit) except Morgan Creek→Tonkawa (WESTEX constraint)

   - 500 kV: 2,000 MVA

3. Cables lookup: Cross-references OSM cables tags from texas_hv_lines.geojson to determine circuit count (cables÷3). Spatial grid lookup finds nearest GeoJSON segment within 8 km.

4. Synthetic transformers: Co-located substations at different voltage tiers within 300 m get a transformer branch (X=0.12 pu, rating = lower-tier rating).

5. Post-processing: Plant outlet stubs and branches < 1 km set to 999,999 MVA (unconstrained). No SPL multiplier, no bridge-load compensation — the topology is meshed enough to not need them.

6. Connected component filter: Keep only the largest connected component.